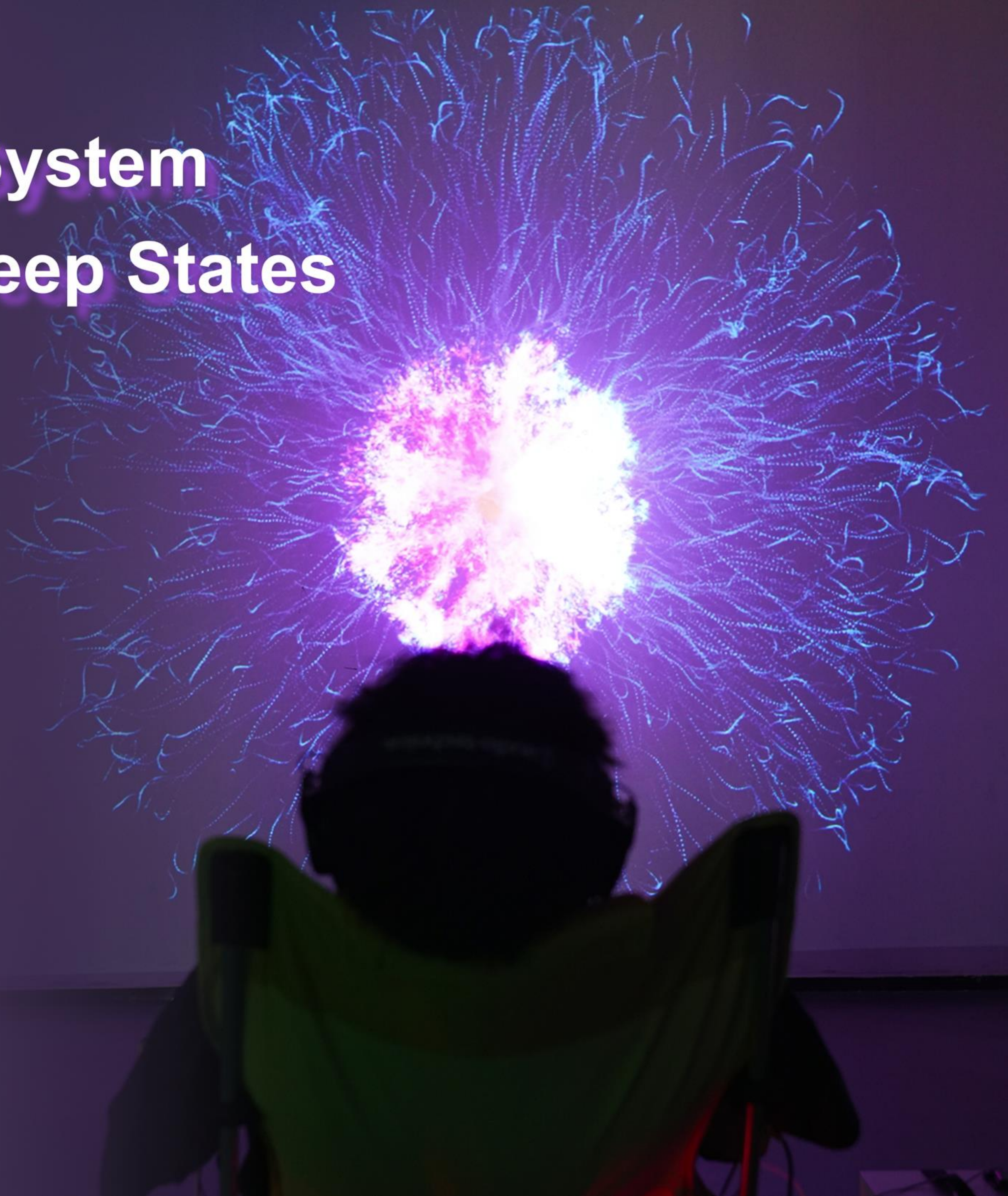
A Multisensory Biofeedback System for Enhancing Positive Pre-Sleep States



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01 | Research Background and Aim

Hyperarousal Model [1]

Somatic activation

- HR ↑, HRV ↓， EDA ↑
- Core temperature ↑
- Cortisol ↑ （HPA axis activation）

Cognitive activation

- Rumination, worry
- Excessive focus on sleep

Cortical activation

- High-frequency EEG activity ↑
- Heightened sensitivity to stimuli

Sleep disorder can be conceptualized as a psychobiological disorder which is linked **not only** on a **psychological level,** but which is associated with measurable **deviations of neuroendocrine and neuroimmunological** variables, as well as **electro- and neurophysiological**, structural and functional alterations of the brain[1].

How to de-arouse in pre-sleep ?

Positive Pre-sleep States

Physiologically

de-arousal is reflected in parasympathetic dominance, lowered HR, and reduced Electrodermal Activity (EDA), indicating diminished sympathetic activation.

Psychologically

it appears as reduced worry and rumination; and emotionally, as a calm and positive mood.

These findings underscore that sleep disorders are **not driven by a single factor**, but by cognitive, somatic, and emotional hyperarousal interaction. To achieve a positive pre-sleep state, it is **necessary to de-arouse in the three dimensions**.

Interoceptive Awareness (IA)

One reasonable pathway to down-regulating hyperarousal is through interoception, which is **the perception of sensations from our body**[2].The Theory of Constructed Emotion [3] conceptualizes emotions as predictive interpretations of bodily signals. Neuroimaging studies further highlight the insula and Anterior Cingulate Cortex (ACC) as central hubs linking IA with higher-order cognitive and affective regulation.

In this pre-sleep state, insufficient IA sustains rumination and hyperarousal, while the capacity to monitor bodily rhythms enables strategies that reduce arousal and promote calm.
By anchoring attention to internal sensations, IA provides a pathway to disengage from cognitive rumination and stabilize emotional states, facilitating the transition to sleep [5].

This suggests that interoceptive processes have the potential to reduce hyperarousal and facilitate a more positive pre-sleep state.

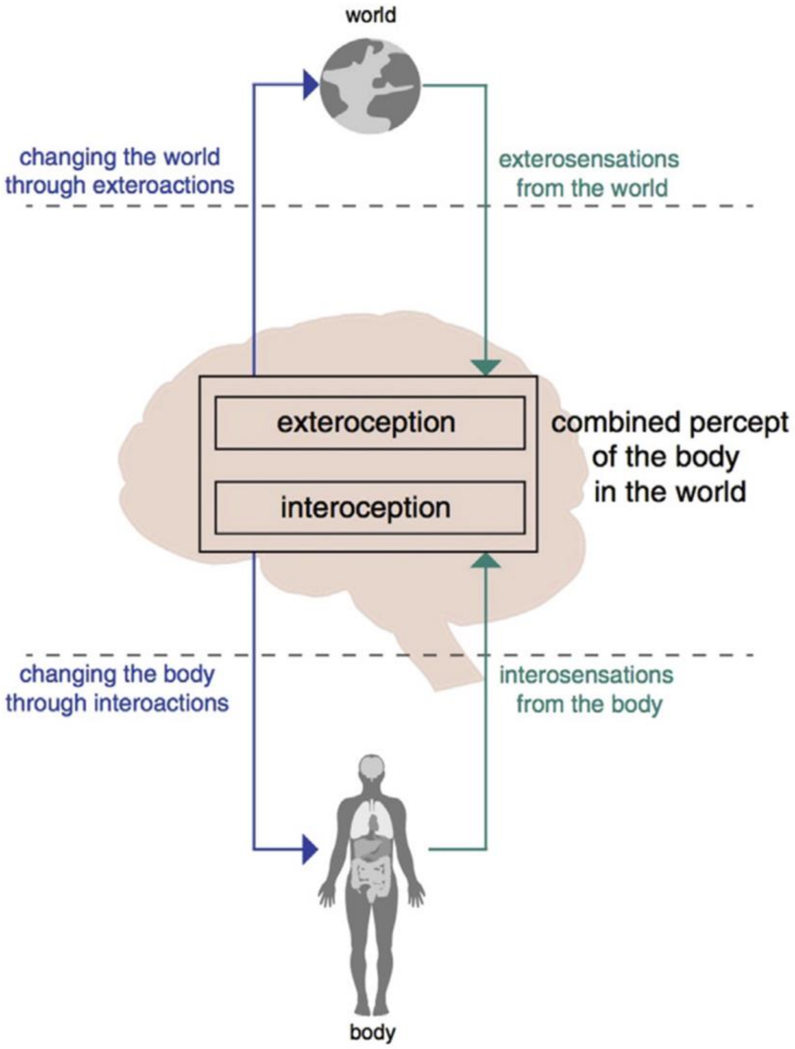


Figure 1 | Highly schematic example of illustrating that both interoceptive information and exteroceptive information are concurrently integrated to inform perceptual representations and action selection with respect to internally directed (e.g., visceromotor, autonomic) and externally directed (e.g., skeletomotor) actions. [4]

Approaches to supporting IA for Mental Health

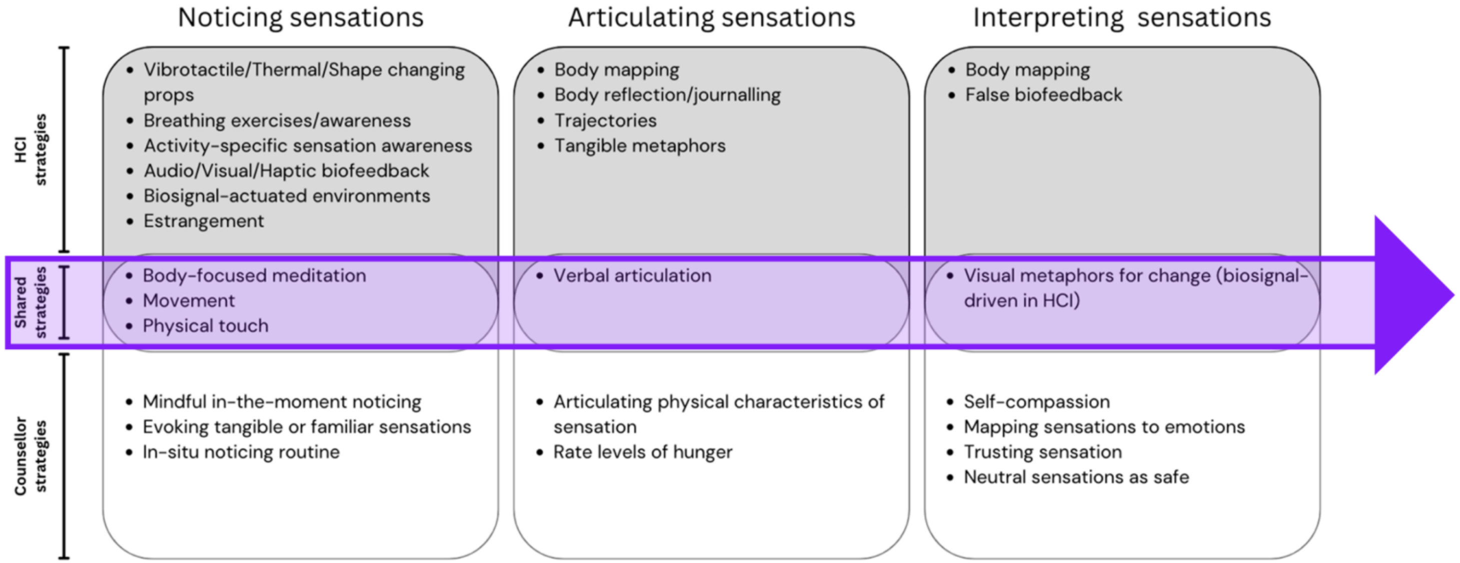


Figure 2 | Strategies used in HCI to cultivate IA (top), strategies used by counselors to cultivate IA (bottom) and overlapping strategies (middle)[6]

The design strategies in both the field of Human-Computer Interaction (HCI) and psychological counseling to support IA can primarily be categorized into three approaches: using technology to assist users in **Noticing, Articulating, and Interpreting Bodily sensations.**

Biofeedback System

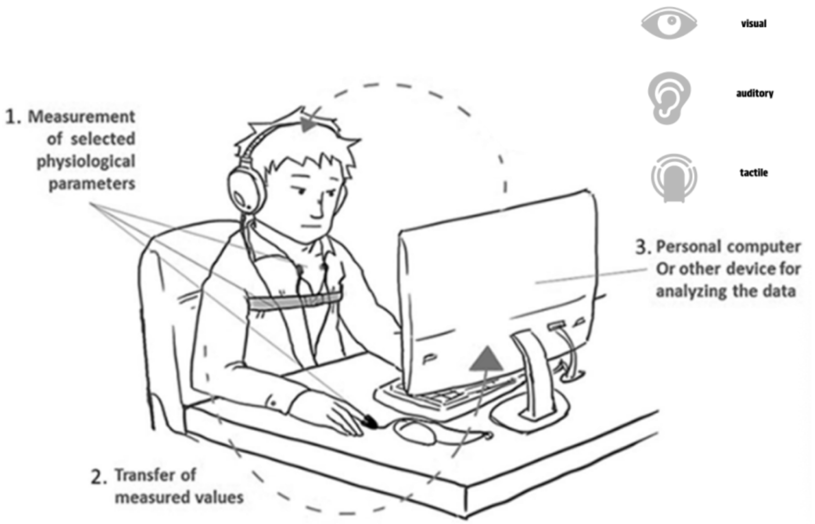


Figure 3 | Biofeedback is a closed-loop system that translates bio-signals into audiovisual displays.[7]

Biofeedback is a powerful tool for stress management and relaxation training. It enables individuals to learn how to regulate their physiological activities in order to restore or maintain autonomic balance [7].

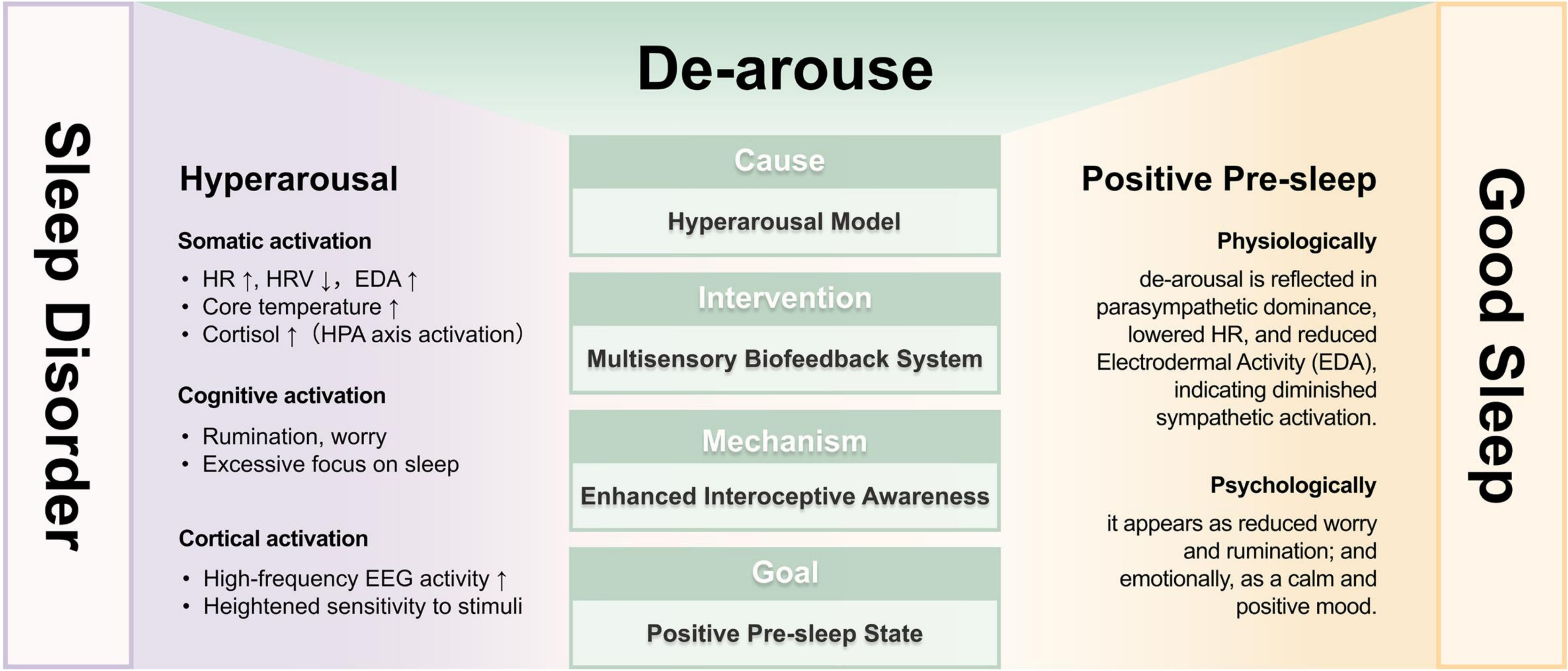
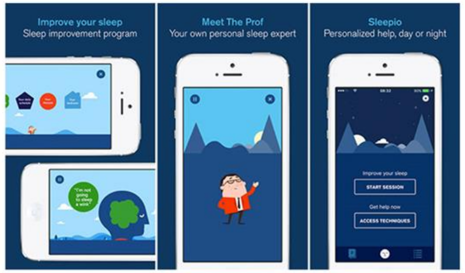


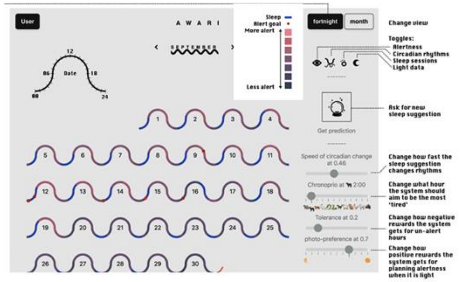
Figure 4 | Theory Frame (self-drawing)

This study aims to address this gap by focusing on the role of biofeedback systems in promoting IA and impacting on arousal levels, highlighting their potential to facilitate a positive transition to sleep.

Sleep Aid Technologies

Based on technical principles and application differences, sleep-assistance technologies can be categorized into four core types: **Wearable Devices Monitoring, Cognitive Behavioral Therapy for Insomnia (CBT-I) , Regulation of Environment , and Drug and Medical Interventions.**




Wearable Devices Monitoring | CBT - I | Regulation of Environment | Passive Biofeedback System

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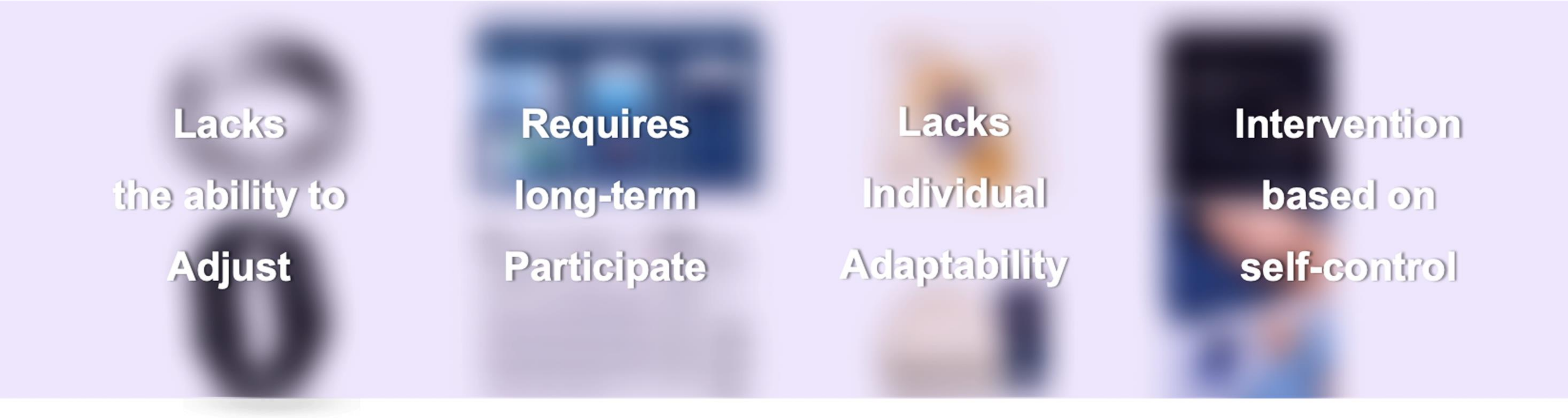
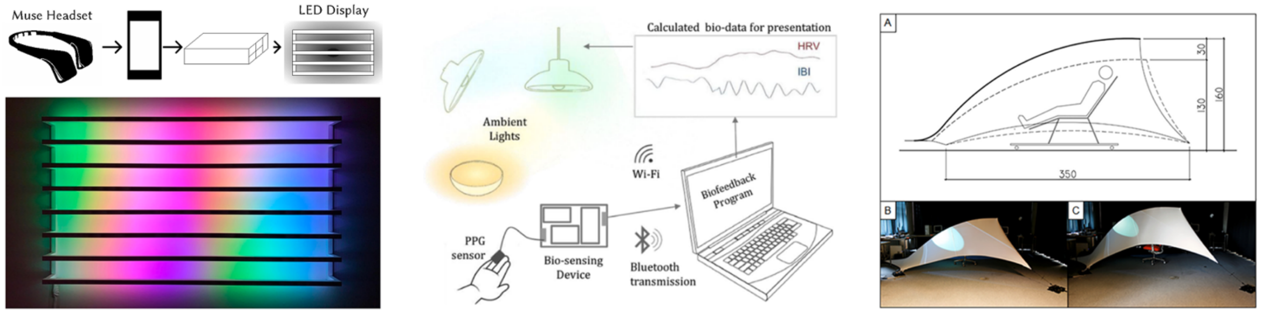


Wearable Devices Monitoring — **CBT - I** — **Regulation of Environment** — **Passive Biofeedback System**

02 | Related Work

Visual biofeedback systems transform physiological signals such as breathing, heart rate, EEG, or heart rate variability (HRV) into light, graphical, or spatial representations.

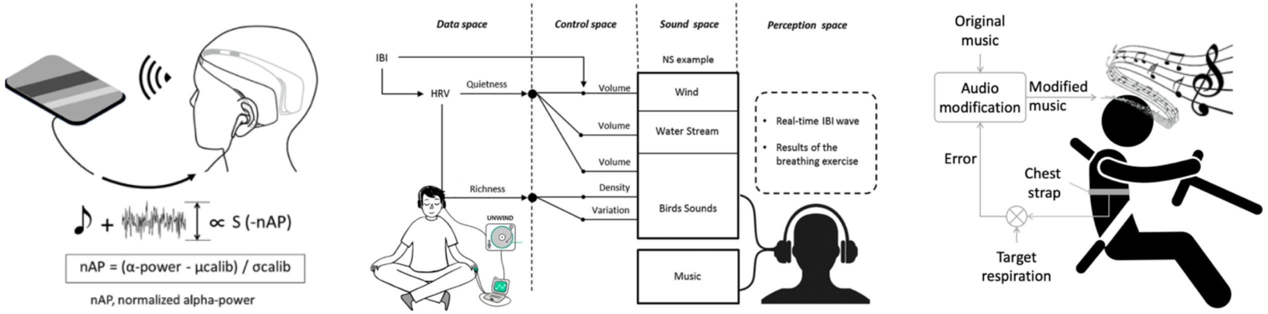
Visual



(a) (b) (c)

Auditory biofeedback systems map physiological signals such as respiration, heart rate, or neural activity onto acoustic parameters—including rhythm, timbre, pitch, and noise intensity

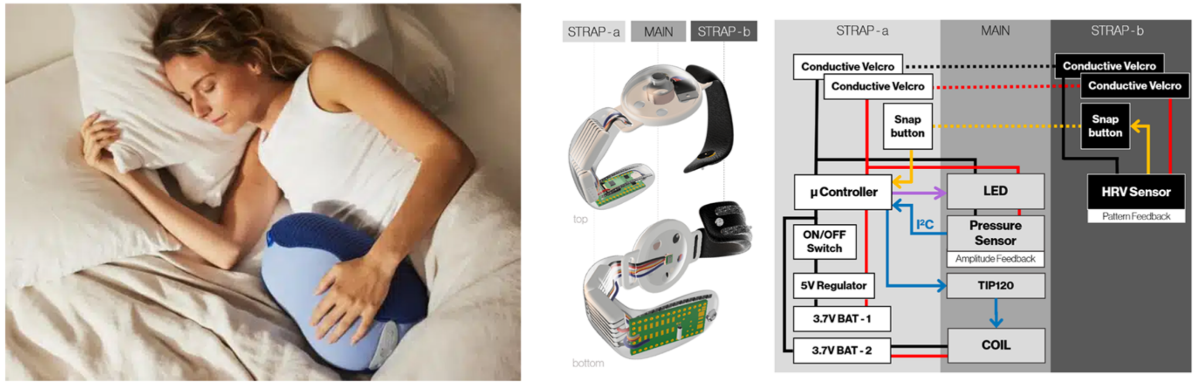
Audio



(a) (b) (c)

Tactile feedback commonly employs metaphorical mappings to represent physiological data: respiration is often conveyed through rhythmic expansion and contraction, while cardiac activity is expressed through temporal interval variations, such as inter-beat intervals (IBI) and heart rate.

Tactile

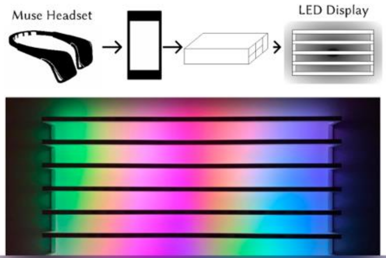


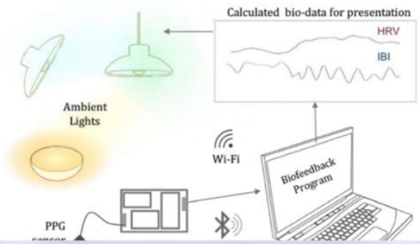
(a) (b)

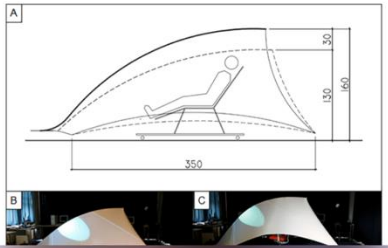
Biofeedback System

Visual biofeedback systems transform physiological signals such as breathing, heart rate, EEG, or heart rate variability

Visual







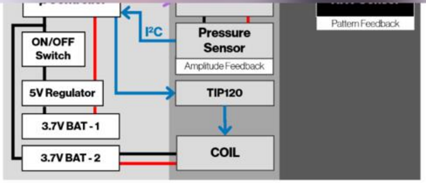
An adaptive biofeedback mechanism that

dynamically shifts modalities based on the user's pre-sleep

through rhythmic expansion and contraction, while cardiac activity is expressed through temporal interval variations, such as inter-beat intervals (IBI) and heart rate.

Tactile

(a)

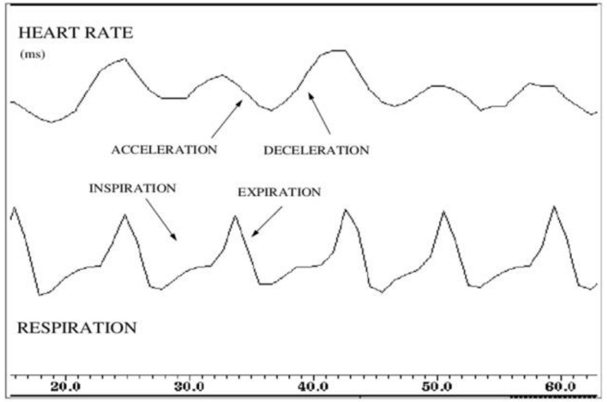


(b)

CRC (Cardiorespiratory Coordination)

The cardiovascular and respiratory systems are **not independent** but are **tightly coupled** through the autonomic nervous system (ANS). This interaction is manifested in the dynamic r**elationship between cardiac and respiratory rhythms**, which can exhibit synchronization or coordination patterns that vary across physiological states.

Physiological Context	CRC Characteristics	HR & Respiration Patterns	Underlying Mechanism
Relaxation / Recovery vs. Stress / Cognitive Load	CRC tends to increase during relaxation and decrease under stress	Relaxation: lower and more stable heart rate with regular breathing; Stress: elevated heart rate and faster, irregular breathing	Shift in autonomic balance: parasympathetic dominance vs. sympathetic activation
Deep Sleep (NREM) vs. REM Sleep	CRC is typically stronger during deep sleep and weaker or less stable during REM sleep	Deep sleep: stable heart rate and respiration; REM sleep: larger fluctuations in heart rate and breathing	Differences in autonomic regulation across sleep stages
Exercise vs. Post-exercise Recovery	CRC patterns change during exercise and gradually strengthen during recovery	Exercise: increased heart rate and breathing rate; Recovery: gradual return to stable rhythms	Coordinated cardiopulmonary regulation for oxygen demand and parasympathetic reactivation
Cardiovascular Disorders vs. Sleep Apnea	CRC may decrease or exhibit abnormal patterns	Reduced heart rate variability or disrupted breathing rhythms	Impaired autonomic regulation or disrupted respiratory–cardiac feedback

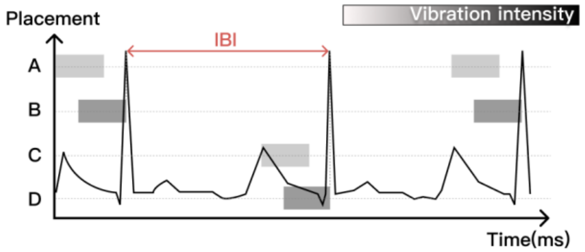


Can making physiological signals perceptible prior to sleep support the process of falling asleep?

Echoes of the Body transforms respiration and heartbeat into synchronized biofeedback to enhance interoceptive awareness and support positive pre-sleep states.

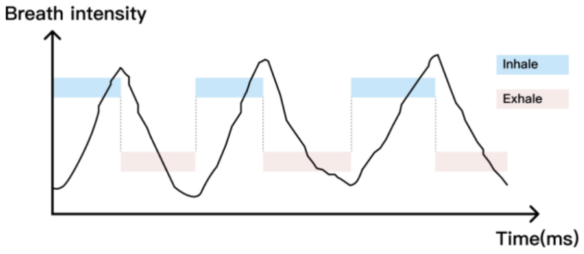
Physical Artifact for Well-being Support

Physical Artifact for Well-being Support（PAWS） represent a class of tangible and haptic technologies that leverage embodied interaction to promote mental well-being[6].



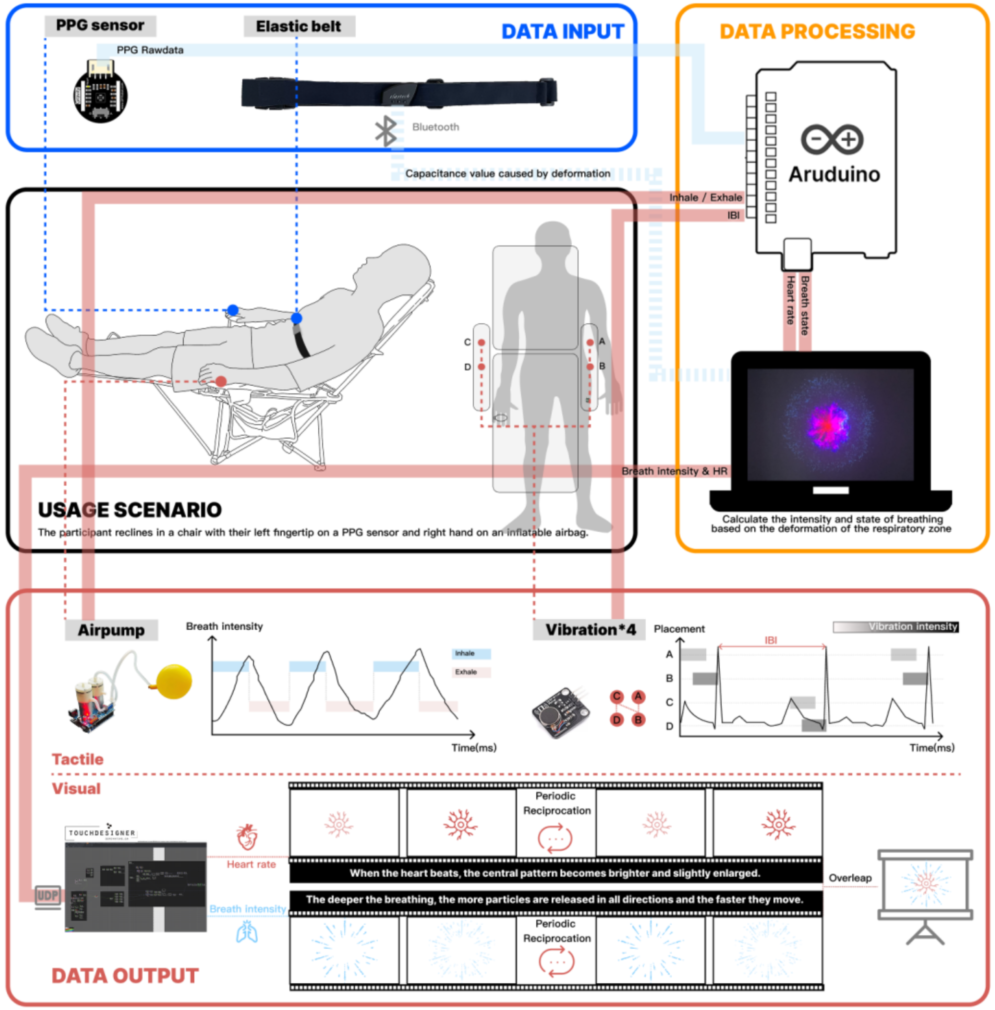
Spatio-Temporal Vibration

A growing body of research is now focusing on exploring the emotional reactions evoked by vibrotactile sensations **structured with spatial and temporal parameters**[7].

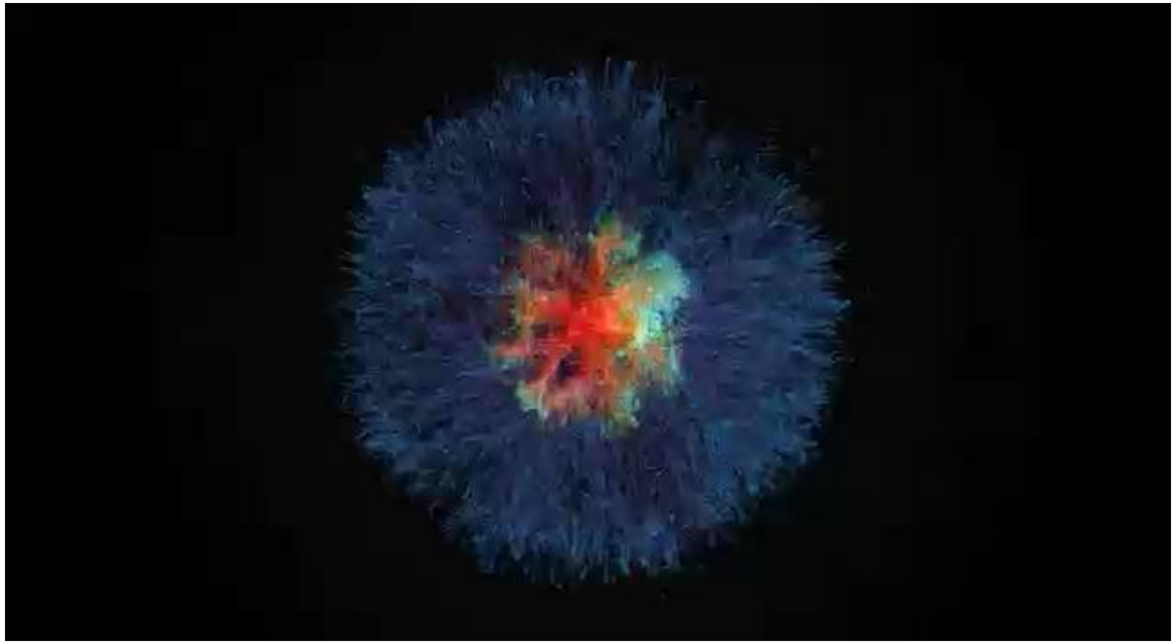


Shape-changing PAWS

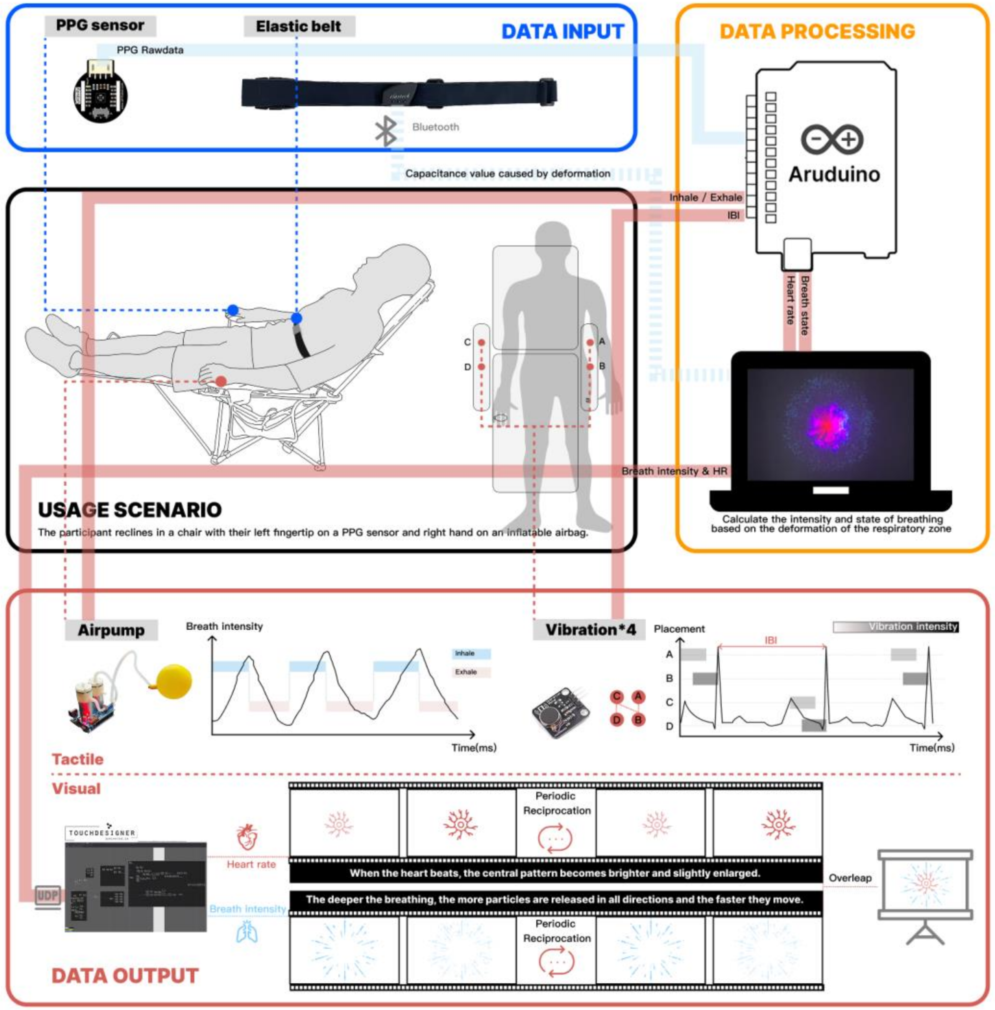
The handheld PAWS was designed as a soft balloon that expands with inhalation and contracts with exhalation, aligning with the body's natural breathing rhythm while avoiding additional cognitive or behavioral activation[9].



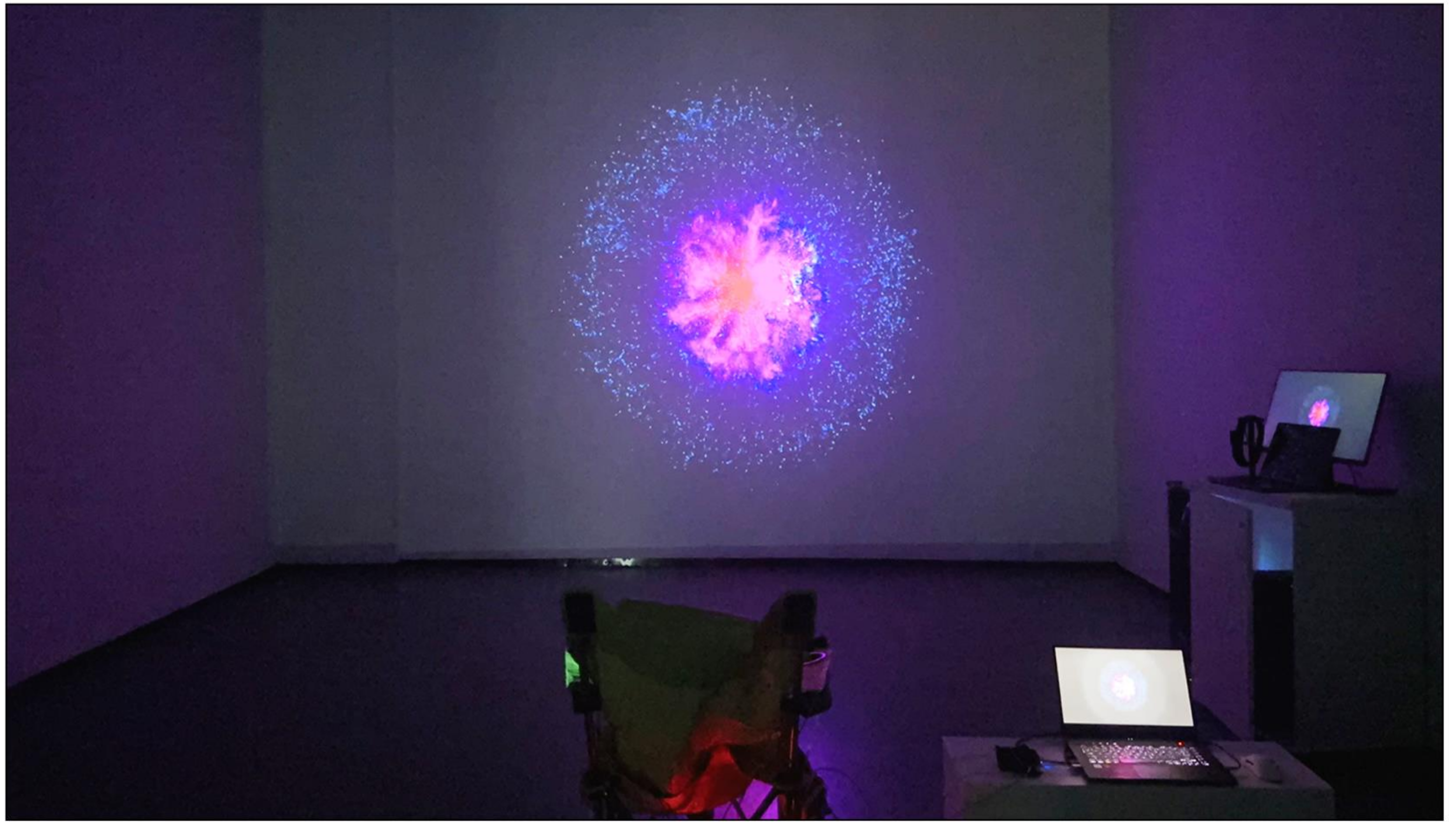
Can making physiological signals perceptible prior to sleep support the process of falling asleep?



We use abstract particle motion to represent breathing and heartbeat, inspired by prior systems [10,11] showing that aesthetic biofeedback helps users better sense their internal states and relax.



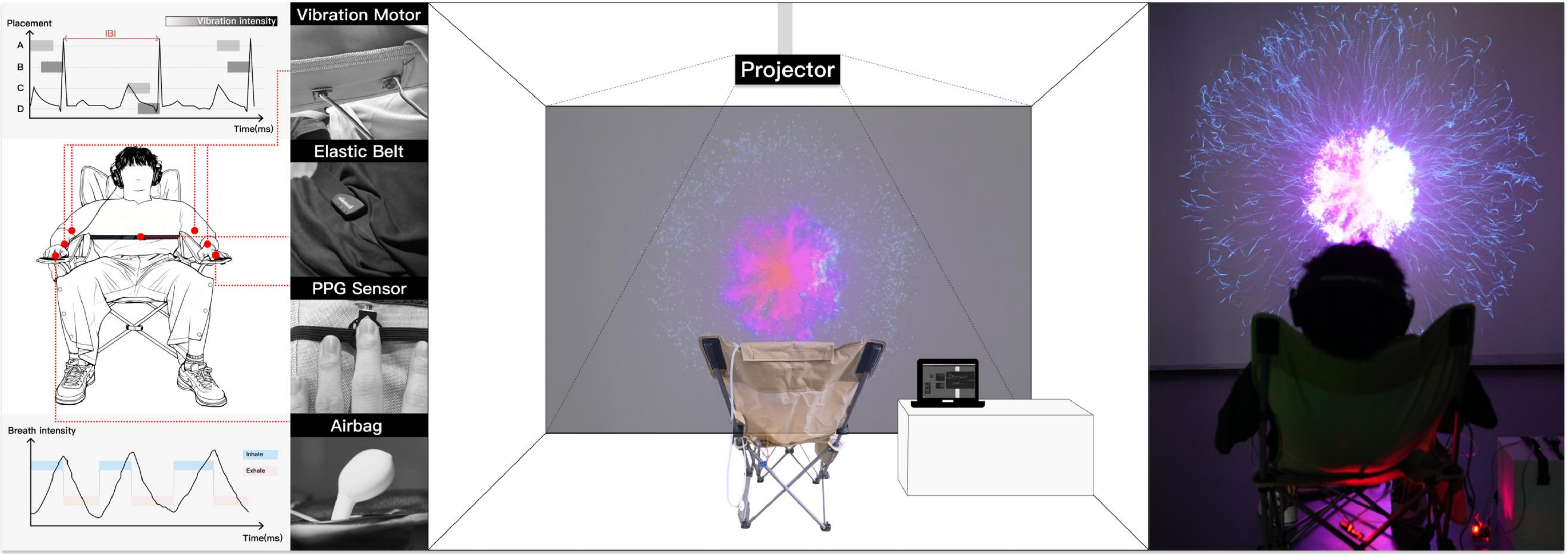
Experimental Design



Experimental Setup

The study was conducted in a quiet, dimly lit room designed to mimic pre-sleep conditions, featuring large-scale wall projection for visual feedback. During the experience, participants wore noise-canceling headphones to eliminate auditory distractions, a respiration belt to record breathing, and placed their hands on the designated areas of the chair's armrests to receive tactile feedback.

Experimental Design



Data Collection

Psychometric Scales

Table 1. Psychometric scales and associated variables used in the study, with brief descriptions.

Category	Scale	Variable	Variable Description
Subjective emotions and stress	VAS-S	Stress	Measures the current level of perceived stress on a visual continuum.
	AS	Valence	Represents the pleasure–displeasure aspect of emotional experience.
		Arousal	Reflects the intensity or activation level of emotional experience.
Interoceptive awareness	Brief MAIA-2	Attention Regulation	Assesses the ability to sustain and control attention toward bodily sensations.
		Emotional Awareness	Evaluates awareness of the connection between bodily sensations and emotional states.
		Self Regulation	Measures the ability to regulate emotional distress through body-based awareness.
Pre-sleep arousal	PSAS	Cognitive Arousal	Captures mental alertness and racing thoughts before falling asleep.
		Somatic Arousal	Captures physical sensations of activation (e.g., rapid heartbeat) experienced before sleep.

Note: Each scale includes one or more variables assessing psychological constructs relevant to the intervention.

Semi-structured Interviews

- What differences did you perceive between the interaction stages, and which did you prefer?
- How did you feel when respiration and heartbeat data were presented simultaneously?
- Do you think the system helped you feel calmer or more relaxed?
- Did the system enhance your awareness of your breathing and heartbeat?
- Did you experience any discomfort, or do you have suggestions for improvement?

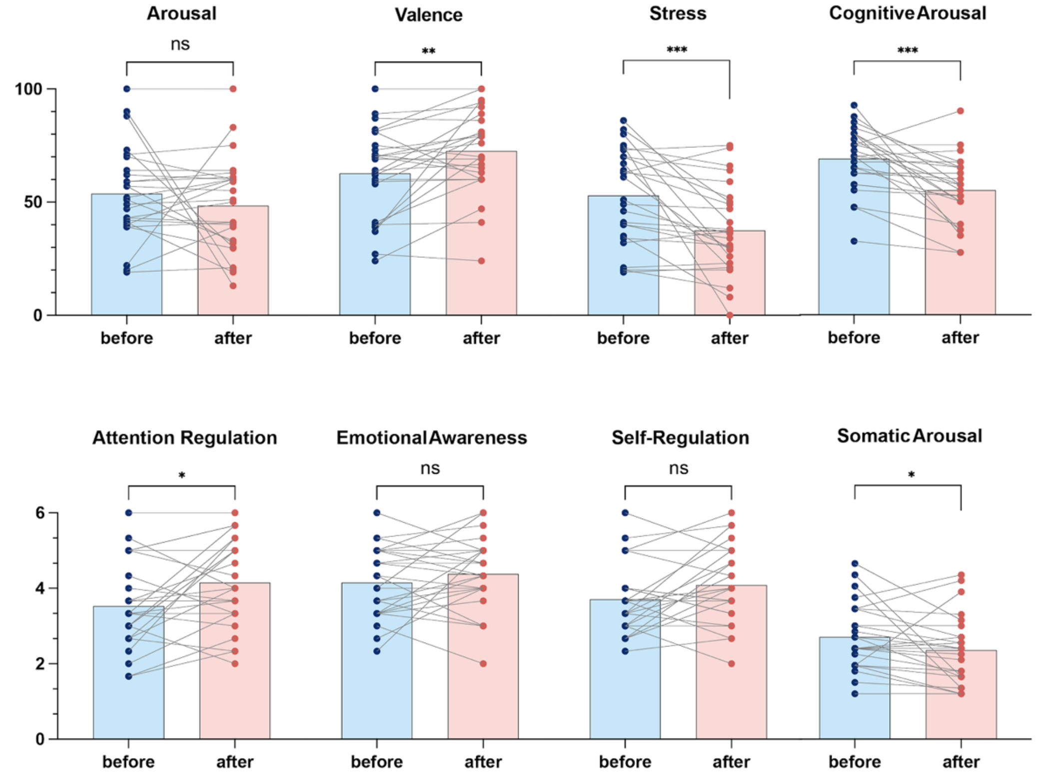
Physiological Monitoring

- **Electrodermal Activity (EDA)**
We recorded participants' electrodermal activity (EDA) during the experience and extracted the tonic and phasic components to evaluate how the interactive system influenced physiological arousal: the tonic component (SCL) reflects baseline arousal, while the phasic component (SCR) captures rapid, stimulus-driven arousal responses[16].

- **Heart Rate (HR)**
We derived heart rate (HR) from PPG signals, and because HR is a key indicator of autonomic nervous system activity and somatic arousal[17], it serves as an essential measure for evaluating whether the system effectively reduces users' arousal levels.

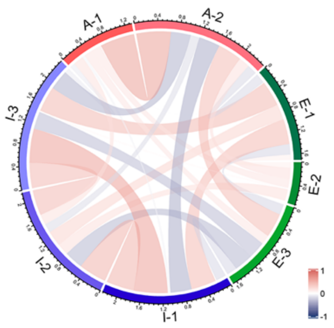
Results —— Psychometric Scales

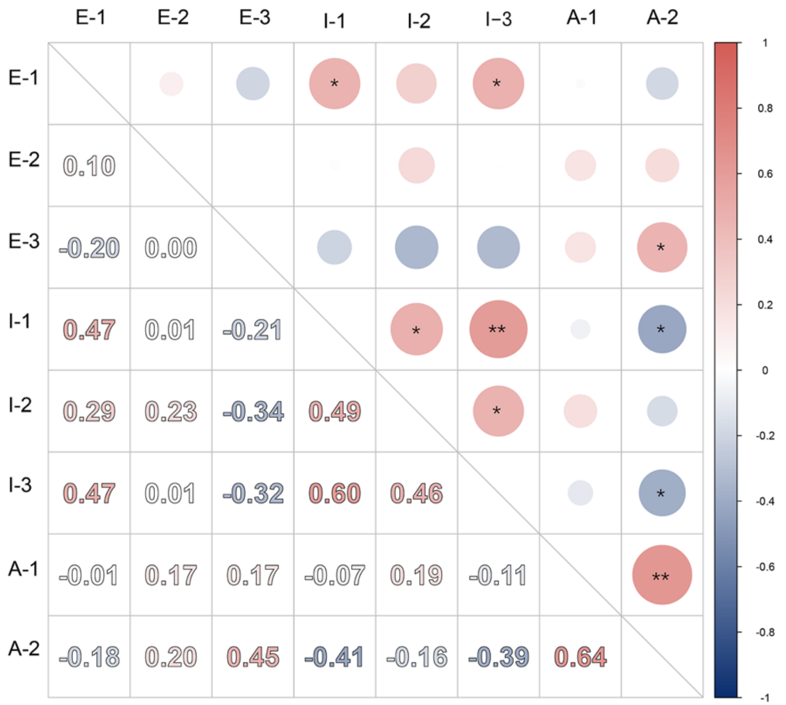
Pre–Post Intervention Analysis



Correlation Analysis

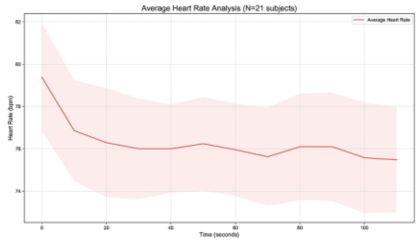
E-1: Valence
E-2: Arousal
E-3: Stress
I-1: Attention Regulation
I-2: Emotional Awareness
I-3: Self Regulation
A-1: Somatic Arousal
A-2: Cognitive Arousal

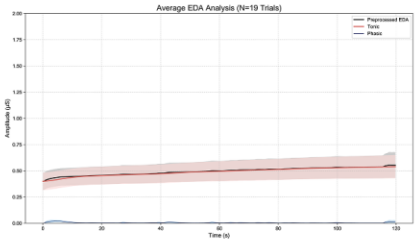




Results

Physiological Monitoring





Result:

- **Heart Rate (HR) ↓**
  Average HR steadily decreased throughout the intervention (N = 21)
  → Indicates a shift toward parasympathetic dominance, meaning increased relaxation.

- **Tonic EDA ↔ (Stable)**
  Tonic skin conductance remained consistent (N = 19)
  → Suggests no rise in baseline sympathetic activity; the system did not increase stress.

- **Phasic EDA ↔ (Stable, No Spikes)**
  Phasic responses showed no abrupt rises or spikes
  → Indicates that the system did not overstimulate or trigger stress-related reactions.

Semi-structured Interviews

Theme 1: Preferences and Perceptions of Feedback Modalities

- ~80% preferred tactile-only feedback for clearer perception of breathing & heartbeat, and stronger immersion. “I closed my eyes, and I felt more peaceful.” (P7)
- Multimodal visual–tactile feedback was engaging but often distracting, with many feeling the two channels were not synchronized (“like two separate systems,” P1).
- Dual-signal input (breathing + heartbeat) felt “competing,” “distracting,” and “cognitively demanding.”
- Respiration feedback was intuitive; heartbeat feedback showed large individual differences in salience.

Our findings show that nearly 80% of participants favored the tactile-only stage, whereas presenting respiration and heartbeat together was frequently experienced as distracting and insufficiently synchronized.

Theme 2: Passivity and self-control

- Visual feedback often triggered active regulation (participants intentionally modified breathing to test visuals). “I would deliberately control my breathing to see the visual change.” (P20)
- This self-control sometimes felt unnatural or effortful.
- Over time, participants shifted to natural breathing, allowing passive reception.
- Tactile-only feedback rarely induced control; instead it supported effortless, intuitive engagement.

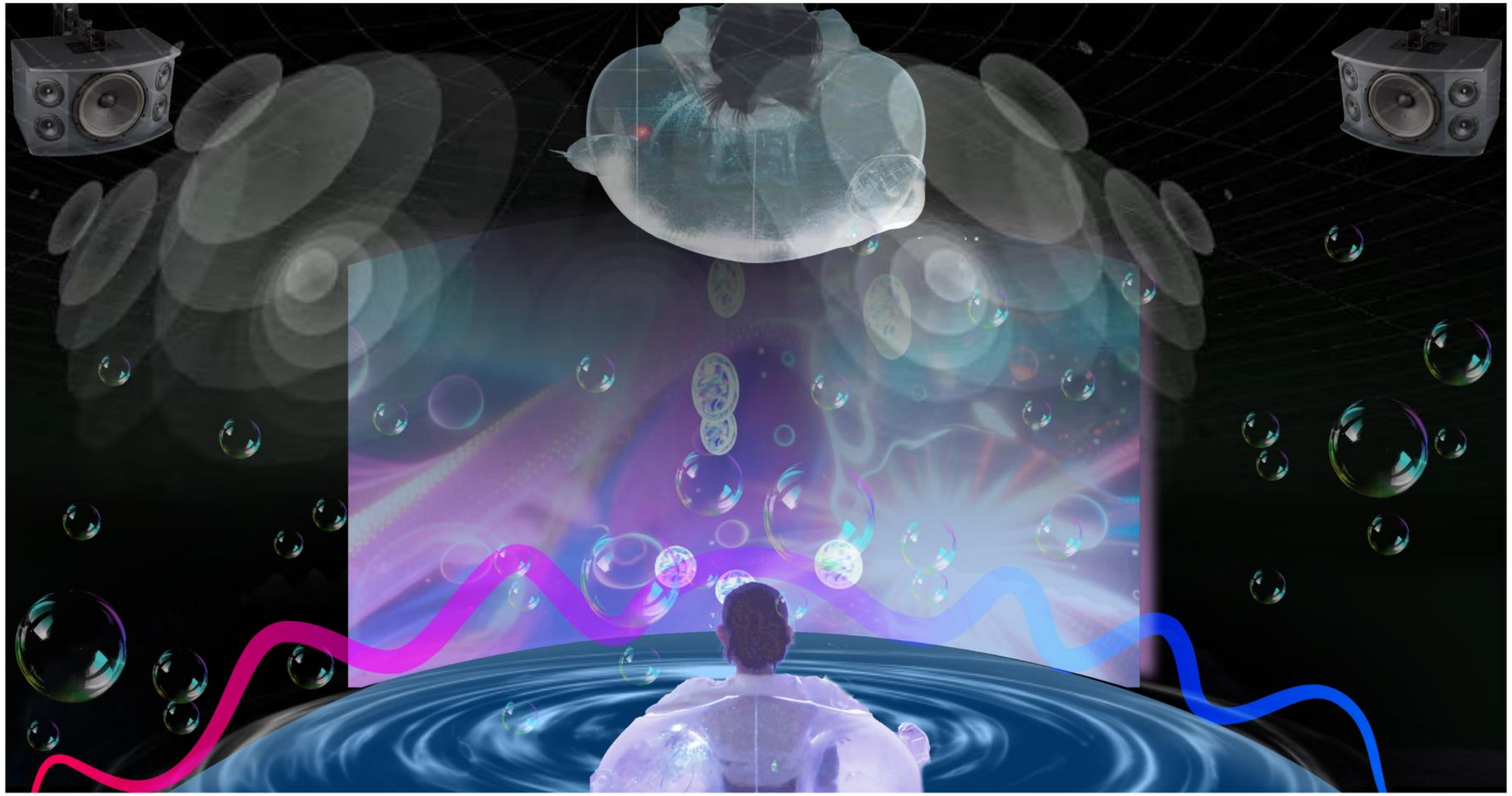
Visual feedback often triggered active self-regulation, while tactile feedback supported effortless, passive engagement better suited for pre-sleep contexts.

Theme 3: Relaxation Effects

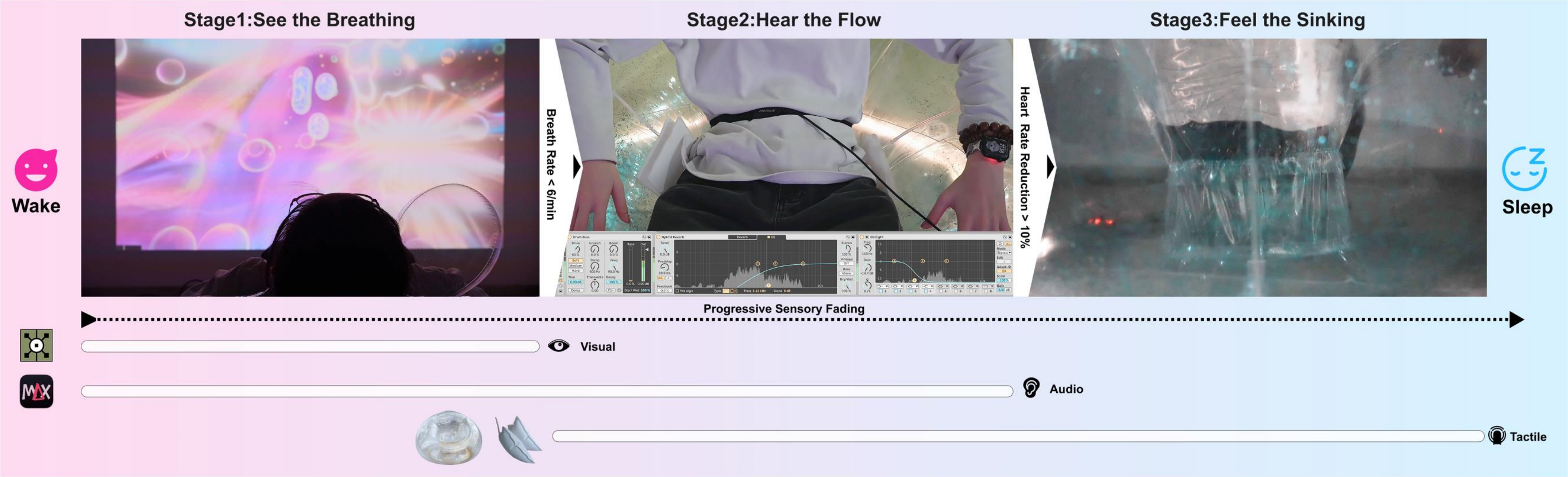
- ~73% participants felt calmer, sleepier, or more relaxed by the end.
- Some reported entering a drowsy, sleep-like state.

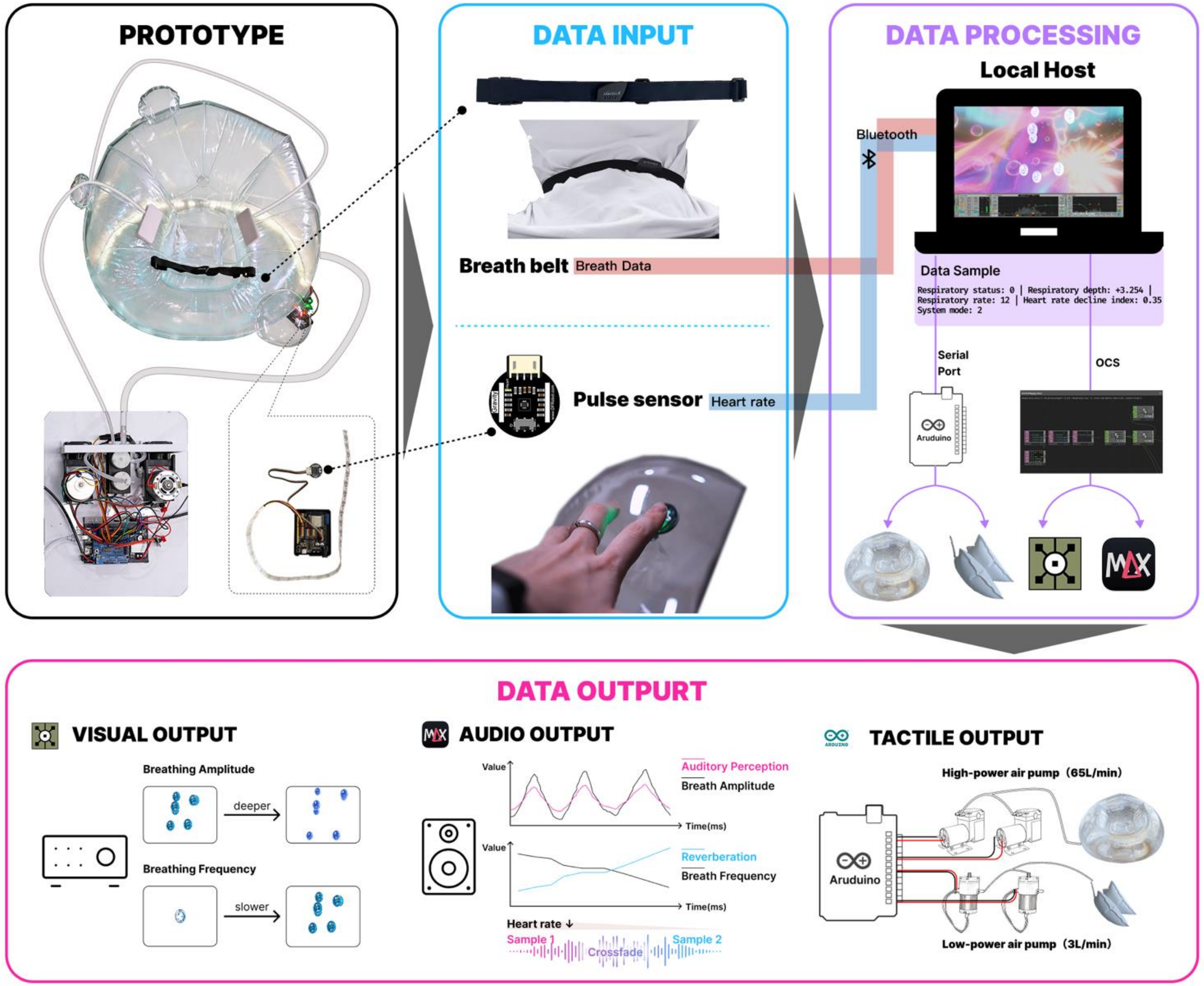
Most participants reported increased calmness and drowsiness, indicating that the system effectively facilitated a transition toward a low-arousal, positive pre-sleep state.

How can interactive systems adapt multisensory biofeedback to support the transition from wakefulness to sleep?

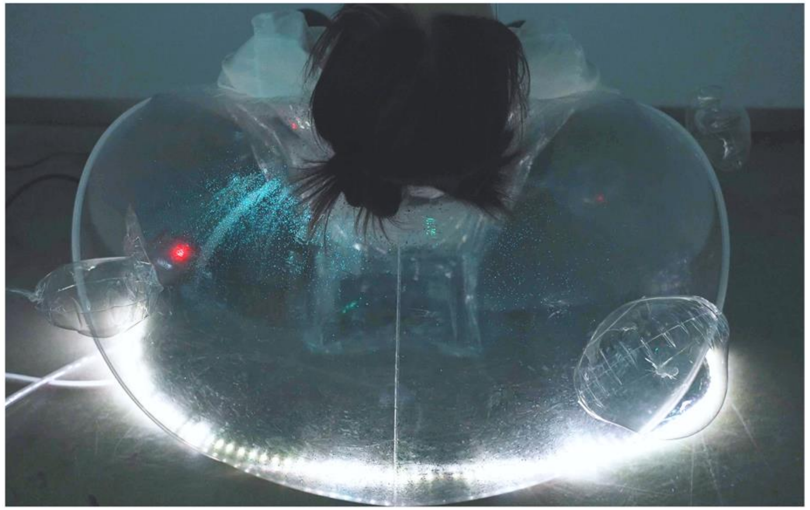


Could biofeedback systems support sleep onset by aligning interactive feedback with the body's sensory gating process through progressive sensory fading?

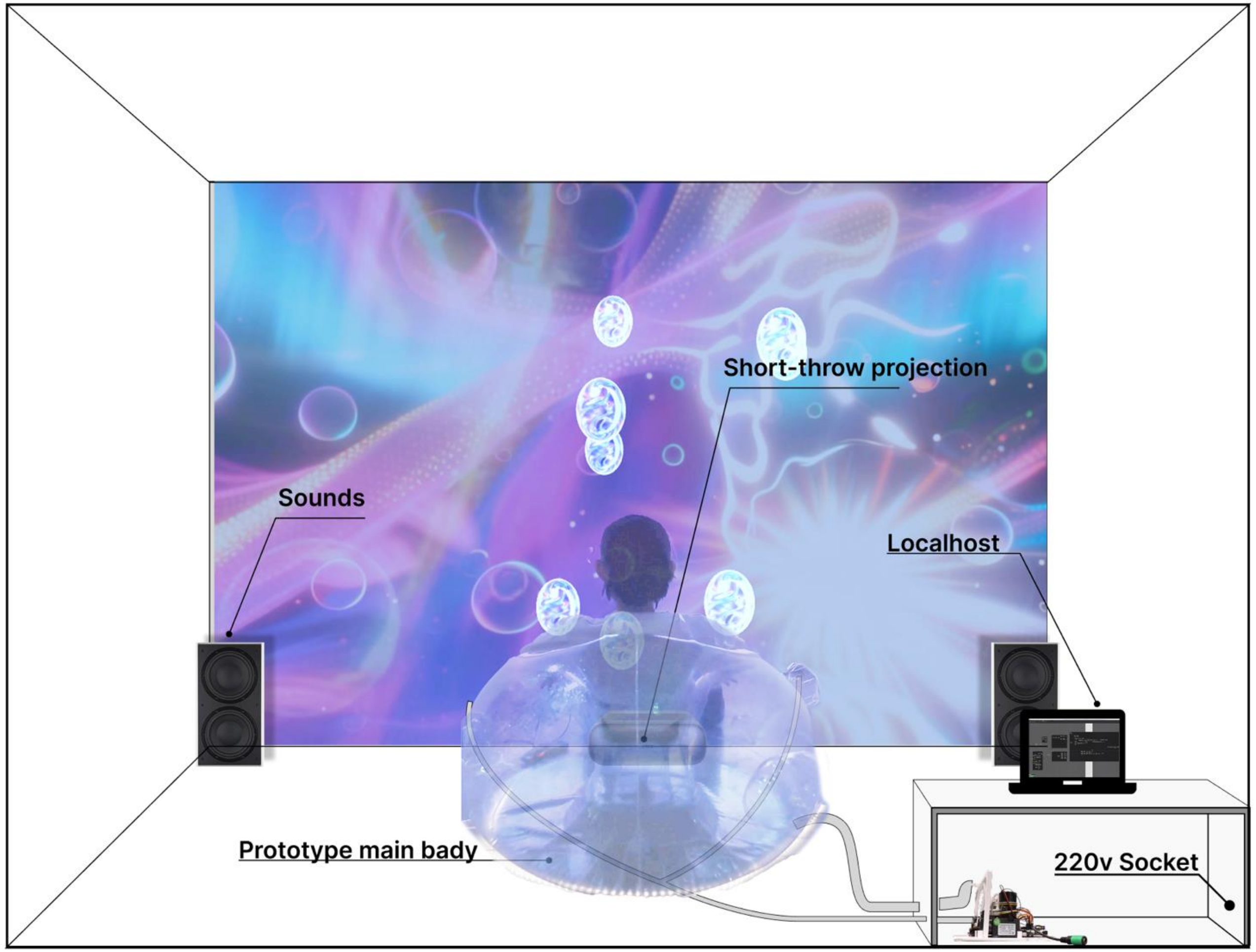


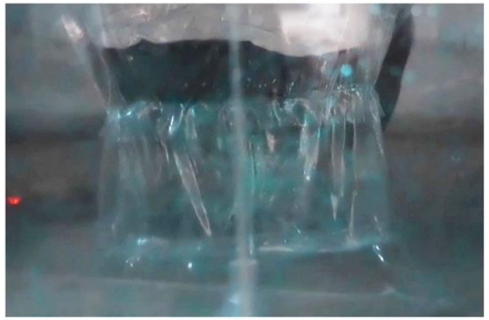

PROTOTYPE
DATA INPUT
Breath belt
Breath Data
Pulse sensor
Heart rate
DATA PROCESSING
Local Host
Bluetooth
Data Sample
Respiratory status: 0 | Respiratory depth: +3.254 | Respiratory rate: 12 | Heart rate decline index: 0.35
System mode: 2
Serial Port
OCS
Aruduino
DATA OUTPURT
VISUAL OUTPUT
Breathing Amplitude
deeper
Breathing Frequency
slower
AUDIO OUTPUT
Value
Auditory Perception
Breath Amplitude
Time(ms)
Reverberation
Breath Frequency
Heart rate
Sample 1
Crossfade
Sample 2
TACTILE OUTPUT
High-power air pump（65L/min)
Aruduino
Low-power air pump（3L/min)

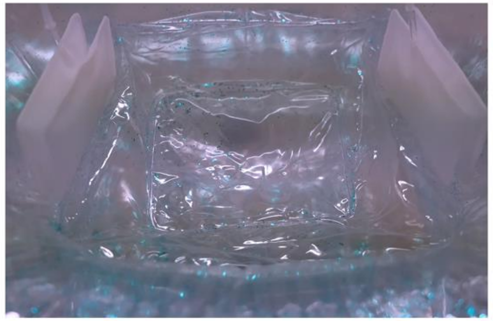

STORM CREW



Technical Specifications and Requirements








Splash into Bubbles
Designing Somatic Immersion
for Sleep Onset by Progressive Sensory Fading

THANKS !

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